Short History of Quantum Mechanics

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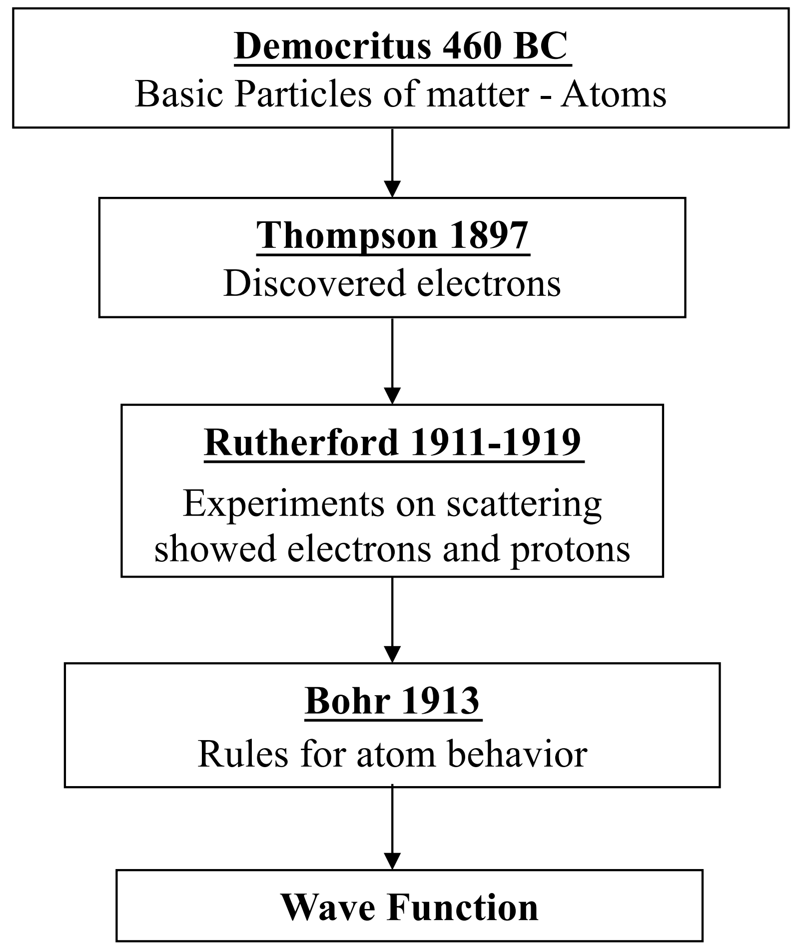
Where did Quantum Mechanics Come From?

- The development of quantum mechanics is strongly tied to the evolving understanding of the nature of matter over many centuries.
- It required a definitive understanding that atoms are composed of separate particles, nuclei and electrons.
- It also required experimental determination of a number of behaviors that were unexplained at the time.

History of Atoms

- The existence of atoms was first proposed in 460 BC.
- But not begun to be understood until the late 19th century.



Historical Development of Quantum Mechanics

- Quantum mechanics arose out of a number of important experimental and conceptual advances.

Waves act like particles

1. Planck theory of black body radiation
2. Photoelectric effect
3. Compton effect

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Particles act like waves

1. De Broglie Postulate
2. Davidson-Germer experiment

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Wave Particle Duality

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Heisenberg Uncertainty Principle

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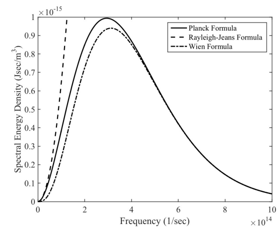
Wave Function

Wave-Particle Duality - Electromagnetic radiation behaves like particles

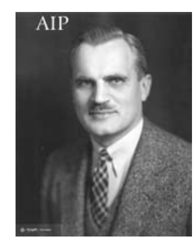
1) Black Body Radiation (Planck 1900) - Oscillators absorb and emit at discrete energies.



$$\varepsilon_n = nh\nu$$

2) Photoelectric Effect (Einstein 1904) - Only the concept of photons, discrete packets of radiation, explain experimental observations.

$$\varepsilon = h\nu$$



3) Compton Effect (1923) - Photons scatter like particles with momentum

$$p = h/\lambda$$

Particle-Wave Duality - Particles can display wave-like behavior.

1) DeBroglie Postulate (1924) - Particles can behave like waves.

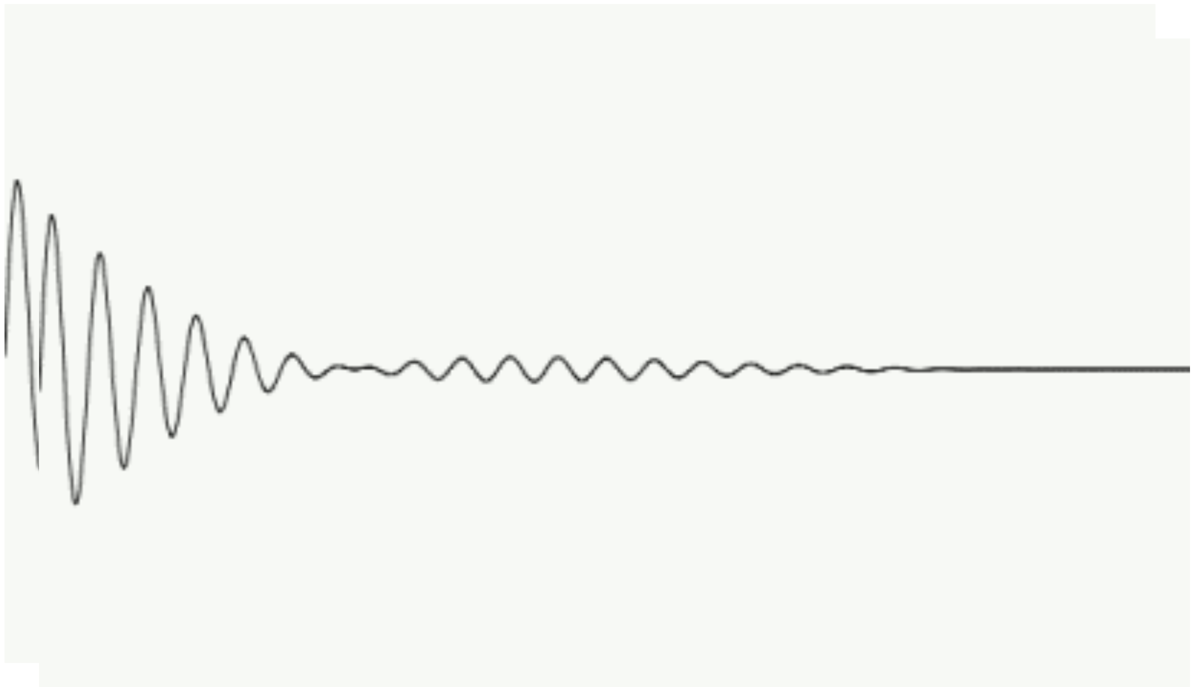
$$\lambda = h/p$$

2) Davisson-Germer Experiment (1927) - Electrons diffracted like waves.

Heisenberg Uncertainty Principle (1927)

Suggested how to deal with duality. Since we can't know everything about a particle, we must content ourselves with a statistical approach. We will describe behavior in terms of a ***wave function***.

Why Would This Work?



http://en.wikipedia.org/wiki/Wave_packet

Schrodinger (1924)

- Used De Broglie's electron wave postulate to develop a wave equation that represents mathematically the distribution of a charge of an electron distributed through space, being spherically symmetric or prominent in certain directions, i.e. directed valence bands, which gave the correct values for spectral lines of the hydrogen atom.